INVESTMENT COMMITTEE DECISION

Cadence Design Systems Decision Memo

TO	Buy-Side Investment Committee	**VALUATION DATE**	September 1, 2026
FROM	Luke Meyer, Junior Financial Analyst	**SUBJECT**	Defer a CDNS position at $313.04

ACTION	MARKET PRICE	BASE DCF	HIGHEST CROSS-CHECK
WATCH / DEFER	**$313.04**	**$83.34**	**$209.47**

Decision and committee ask

I recommend watch/defer with no position in Cadence Design Systems at the September 1, 2026 closing price of $313.04 per share. The integrated FCFF DCF supports $57.20-$111.52, including a base value of $83.34. Synopsys cross-checks are higher at $189.64 from EV/TTM revenue and $209.47 from P/E, but both remain below the market. The methods disagree on the size of the overvaluation, not on the action.

Reopen the decision only when an acquisition-aware DCF and at least two qualified peer observations independently support value at least 15% above the then-current market price. A lower price alone is not enough; the operating evidence must also survive the next two quarterly filings.

Evidence and as-of date

Internal valuation map: three methods, one conclusion



Evidence route	Implied value	Discount	Decision use
Integrated FCFF DCF	$57.20-$111.52	64.4%-81.7%	Primary intrinsic-value range
Synopsys EV/TTM revenue	$189.64	39.4%	Operating-value cross-check
Synopsys P/E	$209.47	33.1%	Equity-value cross-check

I do not average the spread. The DCF prices Cadence's forecast cash flows; the peer methods show what one qualified comparable commands in the market. Synopsys remains imperfect after its Ansys acquisition, so the multiples corroborate the DCF instead of replacing it. All three methods still point to no current entry.

Cadence has real momentum behind it. Recurring revenue represented 80% of FY2025 revenue. In Q2 2026, revenue reached $1.584 billion, backlog hit a record $8.1 billion, and management raised its full-year revenue-growth outlook to roughly 19%. The Semiconductor Industry Association also reported that global semiconductor sales increased 35.1% from Q1 to Q2 2026. These results support near-term demand for Cadence's design software, hardware, and IP. They do not justify carrying boom-level growth into perpetuity.

The base forecast grows revenue from $5.30 billion to $8.44 billion by FY2030, but 74.1% of base enterprise value comes from terminal value. Price requires either a uniform 32.215-point annual growth increase or a 4.871% WACC with base operations. Even an aggressive bounded challenge reaches only $203.95. I do not believe the current boom closes that gap.

External valuation reconciliation: post-date audit only

Post-date benchmark	Value	Method and main driver	Use
ValueInvesting.io | Sep. 11	$137.37	Five-year growth-exit DCF	Lower outside DCF
Fair Value Calculator | Sep. 17	$224.24	Weighted blend of 26 models	Broad-method check
Alpha Spread | retrieved Oct. 18	$272.92	Seven-year FCFE; 34.0% terminal net margin; 29.1x terminal P/E	Optimistic boundary

These later estimates do not enter the September 1 recommendation. Alpha Spread is the optimistic boundary: roughly 10.3% seven-year growth, net margin moving from 35.4% to 34.0%, 101% cash conversion, a 9.3% discount rate, and a 29.1x terminal P/E. Applying those inputs to about $5.838B of starting revenue reproduces $277.46, close to its published $272.92. My five-year, accounting-linked FCFF case uses a 10.0% WACC, 3.0% perpetual growth, and net margins reaching roughly 20%. The gap is mainly optimism about margins and terminal value, not hidden arithmetic.

How WACC and enterprise value were built

The sourced WACC is 10.0%, rounded from 10.0115%. A 4.79% ten-year Treasury yield, 4.23% implied equity risk premium, and 1.2482 industry unlevered beta produce a 1.2751 relevered beta and 10.1837% cost of equity. With 97.2013% equity, 2.7987% debt, a 5.3804% pre-tax debt cost, and a 25.1084% tax rate:

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WACC = 97.2013% x 10.1837% + 2.7987% x 5.3804% x (1 - 25.1084%) = 10.0115%
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The peer bridge is equally reproducible: Synopsys's $85.924B EV divided by $9.416B TTM revenue gives 9.1249x. Applied to Cadence's $5.838B TTM revenue, then bridged through cash, debt, and 275.393M shares, it produces $189.64.

DECISION DISCIPLINE

What must be true before capital moves

Material assumptions and sensitivities

Operating path. Base revenue growth fades from 14% to 5%, gross margin is 86.5%, and operating expenses decline from 67% to 65% of gross profit. These are my analyst assumptions, not company guidance.

Boundaries. Downside and upside remain accounting-linked. The DCF retains FY2025 opening balances; the peer bridge uses June 2026 values. Synopsys remains one qualified peer. The Hexagon layer is separate so filing facts are not blended with assumptions.

Market-price burden of proof

Market-price test	Result	Interpretation
Growth only	+32.215 points/year	FY2026/FY2030 growth becomes 46.2%/37.2% at 10% WACC.
Discount rate only	4.871% WACC	Base operations require a rate 513 bp below the sourced 10.0%.
Bounded challenge	$203.95/share	Still $109.09 below market despite aggressive combined assumptions.
Gross margin only	No solution below 99%	Margin expansion alone cannot reconcile the price.

Cycle regimes and acquisition economics

Decision lens	Operating/risk link	Value and action
AI buildout extends	+3-point growth shift; 87.0% margin; beta 1.35; WACC 10.319%	$93.48 | watch/defer
Orderly normalization	Base operations; beta 1.2751; WACC 10.011%	$83.20 | watch/defer
Cycle reversal	-5-point growth shift; 84.5% margin; beta 1.50; WACC 10.936%	$54.69 | do not initiate
Hexagon D&E overlay	$440M starting revenue; margin/synergy ramp; 10.5% deal WACC	$1.095B acquired EV; combined $78.93/share

The boom case deliberately raises both cash-flow expectations and beta. Hexagon closed February 23, 2026 for $3.101B, including $2.2B cash and 3.224M shares; net consideration after acquired cash was $2.946B. The default overlay values the acquired operation at $1.095B and lowers the combined illustration to $78.93 per share. That result is assumption-dependent, not a purchase-accounting restatement.

The burden of proof

The strongest evidence is not any single target. Rather, it is the consistency across the DCF, peer checks, market-implied dashboard, cycle regimes, and acquisition overlay. For $313.04 to work, several checks would need to be wrong in the same direction.

Evidence I trust now	Remaining uncertainty	Evidence required to reverse
Statements articulate; cash flow ties; scenario order and rate direction pass; the 10.0% WACC is sourced. The separate Hexagon overlay is reproducible.	The overlay is not a full post-acquisition statement rebuild and its operating inputs are analyst assumptions.	Two filings separating acquired and organic growth, margins, integration cash costs, and reinvestment.
Both peer methods remain below market even though they value different claims.	Synopsys is a single peer and its Ansys acquisition changes the comparison.	At least two qualified peers admitted under the same predeclared policy and definitions.
Price requires a 32.215-point annual growth shift or a 4.871% WACC; the bounded challenge reaches only $203.95.	Market-implied tests reveal the burden of proof; they do not prove the required outcomes are impossible.	Two filings showing durable organic growth, margin realization, and cash conversion sufficient to support a revised forecast.

Principal risks, failure conditions, and monitoring triggers

What could change	Monitor and frequency	Committee response
Hexagon integration lifts durable cash flow	Update organic growth, margins, FCFF, cash, debt, and shares within five trading days of each filing.	Rebuild all scenarios and the bridge before reconsidering.
Semiconductor and AI demand accelerates or cools	Track SIA sales, backlog, RPO, guidance, customer R&D commentary, and the industry beta each quarter.	Give the boom credit only when growth converts to FCFF; move toward downside if growth weakens while beta or WACC rises.
Downside becomes more likely	Compare reported growth and gross margin with the 11%-to-2% path and 84.5% downside margin.	Move to do not initiate if weakness persists for two filings.
Model or evidence loses integrity	Re-run statement ties, cash-flow ties, scenario order, source dates, and the terminal-growth test after each update.	Stop if statements do not tie, a bridge is incomplete, or terminal growth reaches WACC.

Action requested and next review point

Approve watch/defer and zero allocation at $313.04. Assign the analyst to refresh the DCF, peer set, and bridge after the next two quarterly filings, with an immediate recheck after a material acquisition update, financing change, or a 100-basis-point move in WACC. Return to committee only with reconciled acquisition-aware cash flows, current peer evidence, and a price that passes the 15% margin-of-safety rule.

Source and validation note. Decision inputs are dated through September 1, 2026. Sources include Cadence and Synopsys filings, Cadence's July 27 release, SIA, dated prices, U.S. Treasury rates, and Damodaran datasets. The later Alpha Spread, ValueInvesting.io, and Fair Value Calculator estimates are a labeled post-date audit, not decision inputs. Full tests and named AI uses appear in Validation and AI Use.

This memo was prepared for FIN 43900 (AI Finance Applications, Purdue) as a learning exercise. It is not investment research or financial advice. The judgments and recommendation remain mine.